EXNO : 4
Date : 3.8.26

Setting up a local Area Network

Aim:

To design, set up, and configure a local Area Network (LAN) Connecting four Computers through a network switch, assign static IP address to all devices, verify connectivity using the ping Command and demonstrate resource / data sharing across the network.

Procedure:

Step ①: Place the Devices

→ open Cisco packet tracer

→ from the device category Pane (select end devices and drag 4 Pcs onto the workspace

→ Select switches and drag 1 Switch (e.g 2960 Series) onto the workspace.

Step ②: Cable the Devices

→ Click Connections → Select Copper Straight - Through Cable.

→ Click each Pc, then click the switch, and select an available fast ethernet port to complete each connection

→ Repeat for all four Pcs (Ports Fa 0/1 to Fa 0/4). Green link lights confirm a successfull physical connection.

Step ③: Assign IP Addresses to Pcs

→ Click a Pc → Desktop tab → IP Configuration.

→ Enter the static IP address and Subnet Mask as per the IP table (section 5) for each Pc.

Step ④: Configure the switch Management IP

→ Configure the switch → CLI tab.

→ Enter the following Commands to assign a management

Enable
Configure terminal
interface vlan 1
ip address 10.1.15 255.0.0.0
no shut down
exit.

Step ⑤: Verify Connectivity using Ping

→ click a Pc → Desktop tab → Command Prompt

→ use the ping Command to test Connectivity to the other Pcs, eg. Ping 10.1.1.2

→ Successfull "Reply from ..." message confirms Connec

Step ⑥: Demonstrate Data sharing

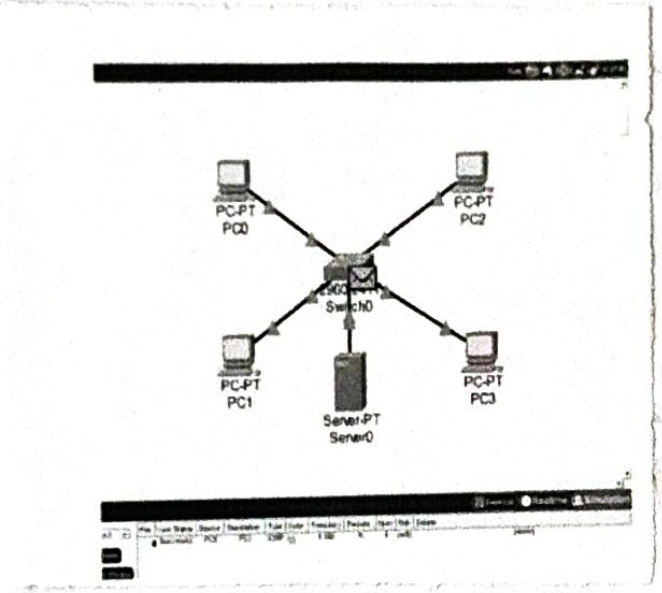
→ Add a server device (end devices → Server PT) and Con it to the switch

→ Assign it an IP address via its Desktop → IP Configura tab.

→ Open the server's Services (Config) tab and enable the HTTP Service

Step ⑦: observe Packet flow in Simulation Mode

→ Switch Simulation : Select a folder → go to properties → click sharing tab → share it with everyone



Result:

Thus the setup & configuration of Lan using Switch and ethernet cables in your lab